

Supplementary Materials

Computational Studies on The Gas Phase Reaction of Methylenimine (CH₂NH) with Water Molecules

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Table S1: Cartesian coordinates (Å) of reactants, complexes, products and transition states obtained using M06-2X/6-311++G(3df,3pd).

(a) For $\text{CH}_2\text{NH} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{NH}_3$

CH₂NH

C	0.05628700	0.58142600	0.00000000
H	-0.84429800	1.19842500	0.00000000
H	1.00853600	1.10768700	0.00000000
N	0.05628700	-0.67872600	0.00000000
H	-0.89596400	-1.04359100	0.00000000

H₂O

H	0.00000000	0.76099200	-0.46653800
O	0.00000000	0.00000000	0.11663500
H	0.00000000	-0.76099200	-0.46653800

CH₂O

C	0.00000000	-0.52447100	0.00000000
H	-0.93797900	-1.10444600	0.00000000
H	0.93797800	-1.10444800	0.00000000
O	0.00000000	0.66946500	0.00000000

NH₃

N	0.00000000	0.00000000	0.11344200
H	0.00000000	0.93915700	-0.26469800
H	-0.81333400	-0.46957900	-0.26469800
H	0.81333400	-0.46957900	-0.26469800

CH₂NH...H₂O

C	-1.25287800	-0.56100700	0.02006900
H	-2.29015400	-0.89421300	-0.02315300
H	-0.47861700	-1.32268900	0.07987300
N	-0.90210100	0.65073100	0.00389800

H	-1.69982300	1.28064600	-0.05584700
H	1.05442400	0.43647600	-0.03364900
O	1.84392400	-0.12431000	-0.07451100
H	2.49475100	0.30519200	0.48116300

CH₂O···NH₃

C	0.00000000	1.10058500	0.00000000
H	0.58057500	1.15742200	0.93373000
H	0.58057500	1.15742200	-0.93373000
N	0.86970200	-1.57900100	0.00000000
H	1.21429200	-2.07357000	-0.81381200
H	1.21429200	-2.07357000	0.81381200
O	-1.19226400	0.99458000	0.00000000
H	-0.13953600	-1.67484600	0.00000000

NH₂CH₂OH-1

C	-0.03408900	0.53637700	0.04810600
H	-0.07386700	1.07583700	0.99620800
H	-0.07886200	1.25906300	-0.76295800
N	1.22008500	-0.15859300	-0.01980500
H	1.35295500	-0.78152900	0.76613300
H	1.28647300	-0.70238700	-0.87072100
O	-1.19277200	-0.26399400	-0.11507500
H	-1.28058700	-0.84714400	0.64193400

NH₂CH₂OH-2

C	0.04291000	0.53866700	0.01615000
H	0.03012400	1.19291300	-0.85997400
H	0.08716800	1.16186200	0.91177900
N	-1.12467700	-0.31425600	-0.06016000

H	-1.34225400	-0.72495800	0.83867400
H	-1.93757000	0.18982900	-0.38714400
O	1.20850400	-0.24093200	0.05448200
H	1.10978500	-0.92440300	-0.61497200

TS1

C	0.33637300	0.67471600	0.02082100
H	0.29127600	1.23488300	0.95406000
H	0.16603700	1.27530300	-0.86564000
N	1.07737800	-0.44447500	-0.12130200
H	1.40511100	-0.77001800	0.78306100
H	-0.21668600	-0.86183800	-0.24843300
O	-1.20368200	-0.18360000	-0.08593900
H	-1.57616600	-0.34649600	0.78864700

TS2

C	-0.03598900	0.54626600	0.04282600
H	-0.14847100	1.12461600	0.96255600
H	-0.01039800	1.24136800	-0.79573100
N	1.19209500	-0.24374900	0.11077200
H	0.95820500	-1.20847200	-0.08967400
H	1.85018300	0.04433400	-0.60086700
O	-1.16909100	-0.26590200	-0.15009100
H	-1.42553000	-0.64598700	0.69208700

TS3

C	0.00000000	0.66943500	0.00000000
H	0.27911700	1.24168300	0.89795900
H	0.27911700	1.24168300	-0.89795900
N	0.89042000	-0.62874600	0.00000000

H	1.44033500	-0.79699100	-0.83597100
H	1.44033500	-0.79699100	0.83597100
O	-1.18311200	0.06227600	0.00000000
H	-0.20694900	-1.00297100	0.00000000

(b) For $CH_2NH+2H_2O \rightarrow CH_2O+NH_3 + H_2O$

$H_2O \cdots CH_2NH \cdots H_2O$

C	-1.52975900	-0.87189400	0.03746700
H	-2.38175400	-1.55242300	0.04421700
O	1.73125600	-1.14756800	0.06775700
H	-0.53170900	-1.30147200	0.09658900
N	-1.65196400	0.38359800	-0.04025500
H	-0.02987700	1.28176900	-0.06966800
H	-2.62566400	0.67544900	-0.09585900
H	1.06776200	2.03595100	0.71490100
O	0.92091100	1.51668700	-0.07709700
H	2.42768000	-1.35935200	-0.55449800
H	1.59852600	-0.18669000	-0.00398000

$CH_2O \cdots NH_3 \cdots H_2O$

C	-0.58315427	-1.24592061	0.42316867
H	-0.35971045	-0.69287744	1.34439000
O	-1.25791583	-0.76991882	-0.45190607
H	-0.18256300	-2.26540001	0.34797628
N	1.85949905	-0.24615528	-0.08983096
H	2.77319922	-0.16141329	0.33794934
H	2.00792497	-0.44683146	-1.07155607
H	1.40656449	0.66311418	-0.03797597

O	-0.40787099	1.81553700	0.14221408
H	-0.96498088	1.09537058	-0.18868036
H	-0.83573032	2.63038316	-0.12209195

NH₂CH₂-OH-1···H₂O

C	-1.08263500	-0.10839200	0.42958500
H	-2.15221400	-0.31885800	0.43291000
O	-0.39086500	-1.20014200	-0.19757300
H	-0.71693400	-0.11558800	1.45299600
N	-0.85817500	1.17900000	-0.13829000
H	0.12897500	1.40764800	-0.15956500
H	-1.25066000	1.27554100	-1.06464500
H	2.78672900	-0.15168800	0.44924200
O	2.02089400	0.21426500	0.00565000
H	-0.71982300	-1.30195900	-1.09398200
H	1.38673600	-0.51073000	-0.09105200

NH₂CH₂-OH-2···H₂O

C	1.07830200	-0.11605600	0.38890700
H	0.76029700	-0.10296800	1.43520300
O	0.56828500	-1.21514100	-0.29570200
H	2.16443000	-0.21435100	0.35299500
N	0.59864100	1.13006500	-0.21981800
H	0.88946100	1.93200800	0.32737600
H	0.99288700	1.23000300	-1.14843400
H	-1.27795200	0.66118300	-0.09029300
O	-1.95443500	0.00615500	0.16472300
H	-0.39518400	-1.19129400	-0.20947600
H	-2.70504000	0.14318800	-0.41424700

TS1...H₂O

C	1.19441400	-0.02949200	0.33615400
H	2.10757700	0.40835400	-0.04965100
O	0.00273400	1.39858400	-0.08738000
H	0.97959000	0.17488600	1.37650300
N	0.70329400	-1.13519800	-0.17266700
H	-0.50096700	-1.06175800	-0.01971300
H	1.03163700	-1.34831900	-1.10528900
H	-2.17086800	-0.51037500	0.75329200
O	-1.59738700	-0.38571600	-0.00344900
H	0.10577500	1.66131400	-1.00606800
H	-0.88505700	0.69628900	-0.03069800

TS2...H₂O

C	1.35620600	-0.18403200	-0.06248600
H	2.04935900	-0.33617600	0.77149800
O	0.33409400	-1.15354100	-0.02398300
H	1.89449000	-0.37256700	-0.99018000
N	0.80125500	1.16432300	-0.07620800
H	-0.13950700	1.16353800	0.30320400
H	1.36199000	1.80033300	0.47421300
H	-2.84207100	0.44783200	-0.49342200
O	-2.14750300	0.12468800	0.08219500
H	-0.04111900	-1.17530600	0.86125000
H	-1.52188600	-0.34289900	-0.48388900

TS3...H₂O

C	1.05290200	-0.32965200	0.28431900
H	1.00644200	-0.26125000	1.38279800
O	0.23480200	-1.20771200	-0.27444200
H	2.09865400	-0.41804000	-0.03024900
N	0.59063000	1.10672200	-0.15705800
H	0.98096600	1.85923800	0.40079700
H	0.81970200	1.24709200	-1.13614300
H	-0.51421500	0.98965200	-0.05009600
O	-1.67609200	0.13436600	0.18684800
H	-0.89494500	-0.70142400	-0.05593800
H	-2.41810700	0.10235700	-0.41692900

Table S2: Harmonic Vibrational frequencies (cm⁻¹) of reactants, complexes, products and transition states obtained using M06-2X/6-311++G(3df,3pd).

(a) For $CH_2NH+H_2O \rightarrow CH_2O+NH_3$

CH ₂ NH	H ₂ O	CH ₂ O	NH ₃	CH ₂ NH...H ₂ O	NH ₂ -CH ₂ -OH -1	NH ₂ -CH ₂ -OH -2	CH ₂ O...NH ₃	TS1	TS2	TS3
1076.99	1619.24	1219.95	1049.9	118.71	281.53	265.5 2	45.79	1900.7 i	342.8	1384i
1122.4	3889.6	1277.9	1661.9	165.9	396.2	352.5	91.0	433.4	504.3	267.3
1169.7	3990.6	1540.6	1661.9	208.1	488.8	539.1	157.4	471.0	845.7	706.8
1370.4		1884.1	3513.4	209.9	814.7	820.9	185.1	666.2	985.0	789.4
1501.1		2959.7	3637.3	371.3	915.0	974.0	298.1	723.1	1079.1	814.0
1745.0		3030.2	3637.3	618.4	1054.9	1031. 3	299.7	764.4	1118.5	1030.9
3068.8		1219.95	1049.9	1081.5	1120.6	1143. 5	1077.7	1028.4	1197.4	1198.5
3153.0		1277.9	1661.9	1132.4	1172.8	1227. 9	1202.8	1082.9	1295.9	1215.2
3471.7		1540.6	1661.9	1158.3	1372.2	1308. 3	1273.5	1254.9	1396.4	1271.9
		1884.1	3513.4	1378.3	1391.0	1391. 1	1539.1	1286.0	1449.8	1377.3
		2959.7	3637.3	1503.0	1433.1	1465. 5	1658.7	1377.4	1545.9	1482.9
				1642.3	1517.5	1543. 1	1663.8	1475.0	1646.3	1543.6
				1749.6	1658.8	1643. 9	1865.7	1552.4	3047.6	1566.0
				3076.3	3062.1	3049. 6	2988.6	1835.0	3097.1	2226.0
				3177.9	3133.0	3089. 3	3059.3	3091.9	3547.4	2970.4
				3496.4	3542.3	3546. 7	3503.4	3188.1	3625.3	2992.1
				3719.1	3627.5	3637. 1	3625.5	3540.7	3915.7	3493.8

(b) For $CH_2NH+2H_2O \rightarrow CH_2O+NH_3+H_2O$

$H_2O \cdots H_2O$	$H_2O \cdots CH_2NH$ $\cdots H_2O$	$OH-CH_2-$ $NH_2-1 \cdots H_2O$	$OH-CH_2-$ $NH_2-2 \cdots H_2O$	$CH_2O \cdots NH_3$ $\cdots H_2O$	$TS1 \cdots H_2O$	$TS2 \cdots H_2O$	$TS3 \cdots H_2O$
159.5	64.38	97.09	118.96	82.46	1326.54i	265.66i	998.22i
172.65	78.61	147.41	187.97	110.44	196.47	114.51	164.17
190.29	164.37	184.65	217.51	130.32	300.91	134.73	351.68
207.41	193.83	209.56	262.63	164.49	407.89	175.64	444.89
378.62	211.46	306.79	291.94	188.35	428.53	226.3	449.77
635.48	219.67	409.76	465.34	217.74	516.47	252.03	531.52
1622.83	233.19	450.31	530.71	221.85	584.94	507.27	658.55
1638.71	240.45	517.21	570.4	250.65	601.3	526.69	717.71
3799.08	412.69	643.77	807.27	361.26	692.45	534.27	769.91
3885.07	441.28	823.96	900.76	420.92	733.15	827.54	920.15
	704.08	919.62	989.08	474.64	783.66	1007.28	1044.28
	843.17	1002.62	1035.87	590.77	1014.5	1056.99	1205.99
	1105.89	1117.34	1146.44	1097.76	1111.73	1128.06	1283.77
	1146.34	1190.06	1237.73	1204.66	1231.86	1250.16	1304.87
	1171.71	1366.43	1323.16	1288.11	1297.47	1262.73	1314.47
	1394.03	1396.55	1439.41	1543.27	1408.88	1403.11	1433.75
	1514.09	1429.18	1473.55	1636.3	1470.59	1457.54	1510.91
	1647.34	1512.95	1537.09	1653.89	1498.39	1553.99	1544.16
	1660.69	1627.88	1626.02	1672.73	1562.66	1612.75	1573.83
	1743.46	1664.09	1642.72	1837.83	1638.95	1650.18	1675.78
	3070.29	3081.15	3053.24	3014	1670.93	3015.26	1869.9
	3185.22	3144.56	3095.6	3101.06	1780.36	3099.67	2149.57
	3485.08	3536.98	3528.12	3475.93	3150.72	3528.02	2983.48
	3500.81	3631.83	3556.14	3598.29	3243.92	3624.75	3043.66
	3647.65	3747.92	3614.44	3636.05	3592.16	3817.49	3502.75
	3952.34	3903.03	3772.36	3732.27	3910.38	3877.89	3593.44
	3970.63	3975.67	3967.35	3975.14	3957.8	3972.26	3962.42

Table S3: Rotational constants (GHZ) of reactants, complexes, products and transition states obtained using M06-2X/6-311++G(3df,3pd).

(a) For $CH_2NH+H_2O \rightarrow CH_2O+NH_3$

CH_2NH	H_2O	CH_2O	NH_3	$CH_2NH \cdots H_2O$	NH_2-CH_2-OH-1	NH_2-CH_2-OH-2	$CH_2O \cdots NH_3$	TS1	TS2	TS3
198.843 79	830.143 41	284.981 50	299.229 96	34.27515	38.68434	39.17952	34.27515	30.314 29	38.518 50	33.334 99
35.3164 0	432.954 16	39.5797 3	299.229 96	5.01425	9.65297	9.95156	5.01425	10.382 26	9.8772 5	11.245 66
29.9899 3	284.549 71	34.7530 5	189.510 96	4.39857	8.66475	8.73185	4.39857	8.5835 8	8.6278 2	9.3532 1

(b) For $CH_2NH+2H_2O \rightarrow CH_2O+NH_3+ H_2O$

$H_2O \cdots H_2O$	$H_2O \cdots CH_2NH \cdots H_2O$	$OH-CH_2-NH_2-1 \cdots H_2O$	$OH-CH_2-NH_2-2 \cdots H_2O$	$CH_2O \cdots NH_3 \cdots H_2O$	TS1 $\cdots H_2O$	TS2 $\cdots H_2O$	TS3 $\cdots H_2O$
214.96040	6.13309	8.84962	8.88416	4.94486	7.77723	9.94246	9.14624
6.51211	3.33696	4.31129	4.60578	4.67846	6.02799	3.88166	5.96131
6.50102	2.18003	3.19148	3.36058	2.63076	3.72609	2.89839	3.98321

Table S4: Electronic energies + ZPE ($E_e + \text{ZPE}$) (Hartree) of reactants, complexes, products and transition states obtained using CCSD(T)/6-311++G(3df,3pd)//M06-2X/6-311++G(3df,3pd).

(a) For $\text{CH}_2\text{NH} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{NH}_3$

Species	ZPE	CCSDT	CCSDT+ZPE
CH_2NH	0.040276	-94.474849	-94.4345734
H_2O	0.021641	-76.337436	-76.3157951
CH_2O	0.027139	-114.335966	-114.308827
NH_3	0.034541	-56.476677	-56.4421364
$\text{CH}_2\text{NH} \cdots \text{H}_2\text{O}$	0.065563	-170.822989	-170.757426
$\text{OH}-\text{CH}_2\text{NH}_2-1$	0.070376	-170.836058	-170.765682
$\text{OH}-\text{CH}_2\text{NH}_2-2$	0.070418	-170.834942	-170.764524
$\text{CH}_2\text{O} \cdots \text{NH}_3$	0.06417	-170.817930	-170.75376
TS1	0.062943	-170.739401	-170.676458
TS2	0.069803	-170.828417	-170.758614
TS3	0.065045	-170.766318	-170.701273

(b) For $\text{CH}_2\text{NH} + 2\text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{NH}_3 + \text{H}_2\text{O}$

Species	ZPE	CCSDT	CCSDT+ZPE
$\text{H}_2\text{O} \cdots \text{H}_2\text{O}$	0.047015	-152.683338	-152.63632
$\text{H}_2\text{O} \cdots \text{CH}_2\text{NH} \cdots \text{H}_2\text{O}$	0.091133	-247.175731	-247.0846
$\text{OH}-\text{CH}_2\text{NH}_2-1 \cdots \text{H}_2\text{O}$	0.09577	-247.185695	-247.08993
$\text{OH}-\text{CH}_2\text{NH}_2-2 \cdots \text{H}_2\text{O}$	0.096576	-247.187795	-247.09122
$\text{CH}_2\text{O} \cdots \text{NH}_3 \cdots \text{H}_2\text{O}$	0.0904	-247.168698	-247.0783
$\text{TS1} \cdots \text{H}_2\text{O}$	0.088363	-247.121489	-247.03313
$\text{TS2} \cdots \text{H}_2\text{O}$	0.094811	-247.174473	-247.07966
$\text{TS3} \cdots \text{H}_2\text{O}$	0.091139	-247.145091	-247.05395

Table S5(a): The T1 diagnostic (computed using CCSD(T)/6 311++G(3df,3pd)) for all species involved in CH₂NH + H₂O and CH₂NH + 2H₂O Reactions.

Species	T1 Diagnostic
CH ₂ NH	0.01228812
H ₂ O	0.00992148
CH ₂ O	0.01542374
NH ₃	0.00828414
CH ₂ NH...H ₂ O	0.01187212
OH-CH ₂ NH ₂ -1	0.01068726
OH-CH ₂ NH ₂ -2	0.01094813
CH ₂ O...NH ₃	0.01373357
TS1	0.01749255
TS2	0.01051416
TS3	0.01475817
H ₂ O...CH ₂ NH...H ₂ O	0.01178066
OH-CH ₂ NH ₂ -1...H ₂ O	0.01081137
OH-CH ₂ NH ₂ -2...H ₂ O	0.0110812
CH ₂ O...NH ₃ ...H ₂ O	0.01345468
TS1...H ₂ O	0.01635814
TS2...H ₂ O	0.01055299
TS3...H ₂ O	0.01310301

Table S5(b): The T1 diagnostic (computed using CCSD(T)/6 311++G(3df,3pd)) for all “Trial” TSs and Complex

Species	T1-Diagonistic
Complex	0.01187212
4	0.01152359
4.2	0.0115008
4.4	0.01148309
4.6	0.01146862
4.8	0.01145654
5	0.0114467
5.2	0.01143887
5.4	0.01143272
5.6	0.0114278
5.8	0.01142384
6	0.0114206
6.2	0.01141789
6.4	0.01141567
6.6	0.01141383
6.8	0.01141228
7	0.01141228
7.2	0.01140985
7.4	0.0114089
7.6	0.01140736
7.8	0.01140671
8	0.01542374

Table S6: Calculated equilibrium constant ($\text{cm}^3 \text{ molecule}^{-1}$) for the formation of two-body and three-body complex.

Temp	Ke ($\text{H}_2\text{O} \cdots \text{H}_2\text{O}$)	Ke($\text{CH}_2\text{NH} \cdots \text{H}_2\text{O}$)	Ke($\text{H}_2\text{O} \cdots \text{H}_2\text{O} + \text{CH}_2\text{NH}$)	Ke($\text{CH}_2\text{NH} \cdots \text{H}_2\text{O} + \text{H}_2\text{O}$)
200	3.4E-21	1.0E-20	8.3E-18	3.6E-18
225	1.4E-21	2.7E-21	7.0E-19	4.4E-19
250	6.9E-22	9.9E-22	9.7E-20	8.3E-20
275	4.0E-22	4.3E-22	2.0E-20	2.2E-20
300	2.6E-22	2.2E-22	5.2E-21	7.1E-21
325	1.8E-22	1.3E-22	1.7E-21	2.8E-21
350	1.3E-22	8.0E-23	6.7E-22	1.3E-21
375	1.0E-22	5.4E-23	3.0E-22	6.4E-22
400	8.3E-23	3.9E-23	1.5E-22	3.6E-22
500	4.8E-23	1.6E-23	1.9E-23	6.6E-23
600	3.6E-23	9.1E-24	5.4E-24	2.3E-23
700	3.1E-23	6.7E-24	2.3E-24	1.2E-23
800	3.0E-23	5.5E-24	1.3E-24	7.2E-24
900	2.9E-23	5.0E-24	8.3E-25	5.2E-24
1000	3.0E-23	4.7E-24	6.1E-25	4.1E-24
1100	3.2E-23	4.7E-24	4.9E-25	3.5E-24
1200	3.4E-23	4.7E-24	4.1E-25	3.1E-24
1300	3.7E-23	4.8E-24	3.6E-25	2.9E-24
1400	4.0E-23	5.0E-24	3.3E-25	2.7E-24
1500	4.3E-23	5.2E-24	3.1E-25	2.6E-24
1600	4.7E-23	5.5E-24	3.0E-25	2.6E-24
1700	5.1E-23	5.9E-24	2.9E-25	2.6E-24
1800	5.5E-23	6.2E-24	2.9E-25	2.6E-24
1900	6.0E-23	6.7E-24	2.9E-25	2.7E-24
2000	6.6E-23	7.1E-24	2.9E-25	2.7E-24

Table S7: Capture Rate constants for the CH₂NH +H₂O Entrance Channel using CVTST and μ CVTST methods.

T (K)	Canonical Approach		Microcanonical Approach	
	Dissociation (s ⁻¹)	Association (cm ³ molecule ⁻¹ s ⁻¹)	Dissociation (s ⁻¹)	Association (cm ³ molecule ⁻¹ s ⁻¹)
200	8.72E+08	8.78E-12	8.57E+08	8.62E-12
225	3.97E+09	1.09E-11	3.77E+09	1.04E-11
250	1.35E+10	1.33E-11	1.24E+10	1.22E-11
275	3.71E+10	1.61E-11	3.29E+10	1.43E-11
300	8.42E+10	1.87E-11	7.39E+10	1.64E-11
325	1.62E+11	2.07E-11	1.47E+11	1.87E-11
350	2.85E+11	2.30E-11	2.63E+11	2.12E-11
375	4.67E+11	2.54E-11	4.36E+11	2.37E-11
400	7.20E+11	2.81E-11	6.77E+11	2.64E-11

Table S8: (a) Tunneling correction for $\text{PRC} \rightarrow \text{TS} \rightarrow \text{CH}_2\text{NH}_2\text{OH}$.

Temp	Tunneling
500	5.5
600	2.6
700	2.0
800	1.6
900	1.5
1000	1.4
1100	1.3
1200	1.2
1300	1.2
1400	1.1
1500	1.1
1600	1.1
1700	1.1
1800	1.1
1900	1.0
2000	1.0

Table S8: (b) Tunneling correction for $\text{PRC}_w \rightarrow \text{TS}_w \rightarrow \text{CH}_2\text{NH}_2\text{OH}_w$

Temp	Tunneling
500	2.9
600	1.9
700	1.6
800	1.4
900	1.3

1000	1.2
1100	1.2
1200	1.2
1300	1.1
1400	1.1
1500	1.1
1600	1.1
1700	1.1
1800	1.1
1900	1.1
2000	1.1

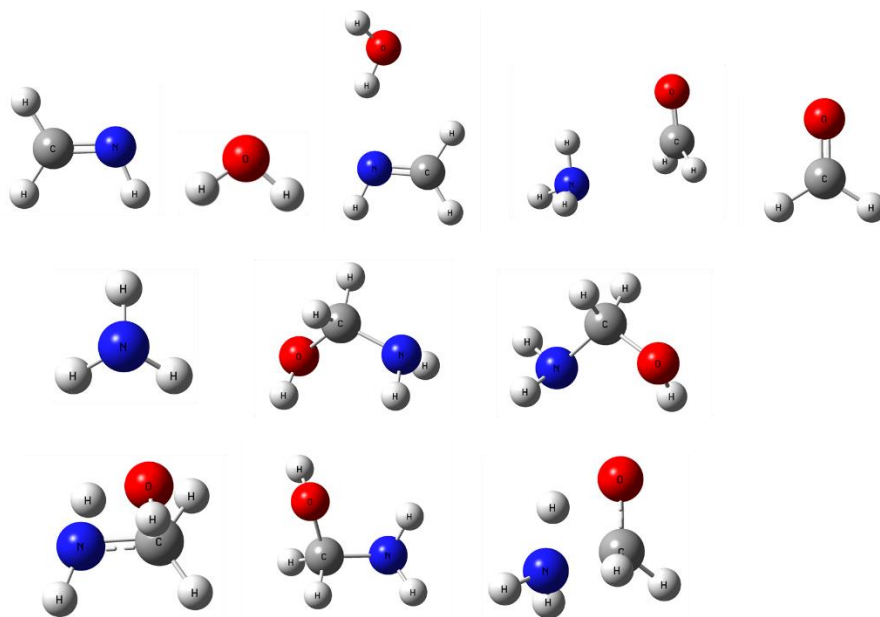
Table S9 (a): Unprojected Harmonic Vibrational Frequencies of some of the Trail TS.

			Un-Projected		
Bond Length (Å)	4	5	6	7	8
	-65.08	-32.34	-18.82	-10.71	-7.23
	58.21	58.31	50.27	43.35	44.15
	65.12	84.28	88.18	85.96	84.16
	126.6	152.17	108.95	138.85	94.21
	203.56	181.89	173.37	169.01	172.24
	378.56	241.75	211.08	193.15	200.74
	1084.84	1083.48	1083.23	1083.07	1083.3
	1120.57	1123.65	1123.82	1123.15	1123.07
	1168.74	1171.75	1172.89	1173.14	1173.03
	1377.86	1375.7	1375.02	1374.6	1374.89
	1503.01	1504.32	1504.86	1504.95	1504.91
	1636.83	1630.82	1627.51	1628.38	1628.4
	1754.74	1754.3	1753.75	1753.32	1753.81
	3063.24	3066.66	3062.3	3059.52	3058.7
	3158.03	3165.15	3167.36	3167.87	3167.44
	3487.62	3470.91	3465.21	3463.93	3462.77
	3867.21	3889.38	3898.3	3901.61	3899.33
	3974.25	3991.59	3997.42	4001.27	3999.56

Table S9 (b): Projected Harmonic Vibrational Frequencies of some of the Trail TS

		<i>Projected</i>			
<i>Bond Length (Å)</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
	57.3	55.8	48.5	42.8	44.1
	63.0	81.8	82.6	75.3	65.8
	126.5	151.7	108.4	138.6	94.2
	203.1	181.9	173.4	168.5	172.2
	377.9	241.5	211.1	193.1	199.3
	1084.4	1082.6	1082.5	1048.2	993.4
	1120.5	1122.0	1107.7	1084.6	1083.3
	1168.7	1171.7	1172.1	1170.2	1169.0
	1306.7	1303.7	1308.4	1307.5	1342.5
	1494.9	1496.4	1499.5	1501.1	1504.8
	1625.4	1616.3	1608.1	1622.4	1602.6
	1723.8	1722.2	1724.7	1723.0	1734.2
	3063.2	3066.5	3061.8	3059.0	3055.7
	3158.0	3165.1	3167.3	3167.9	3166.1
	3465.0	3447.0	3434.4	3412.9	3410.1
	3835.4	3858.1	3868.8	3832.3	3863.2
	3952.6	3969.6	3976.9	3953.4	3981.3

(a) For $\text{CH}_2\text{NH} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{NH}_3$



(b) For $\text{CH}_2\text{NH} + 2\text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{NH}_3 + \text{H}_2\text{O}$

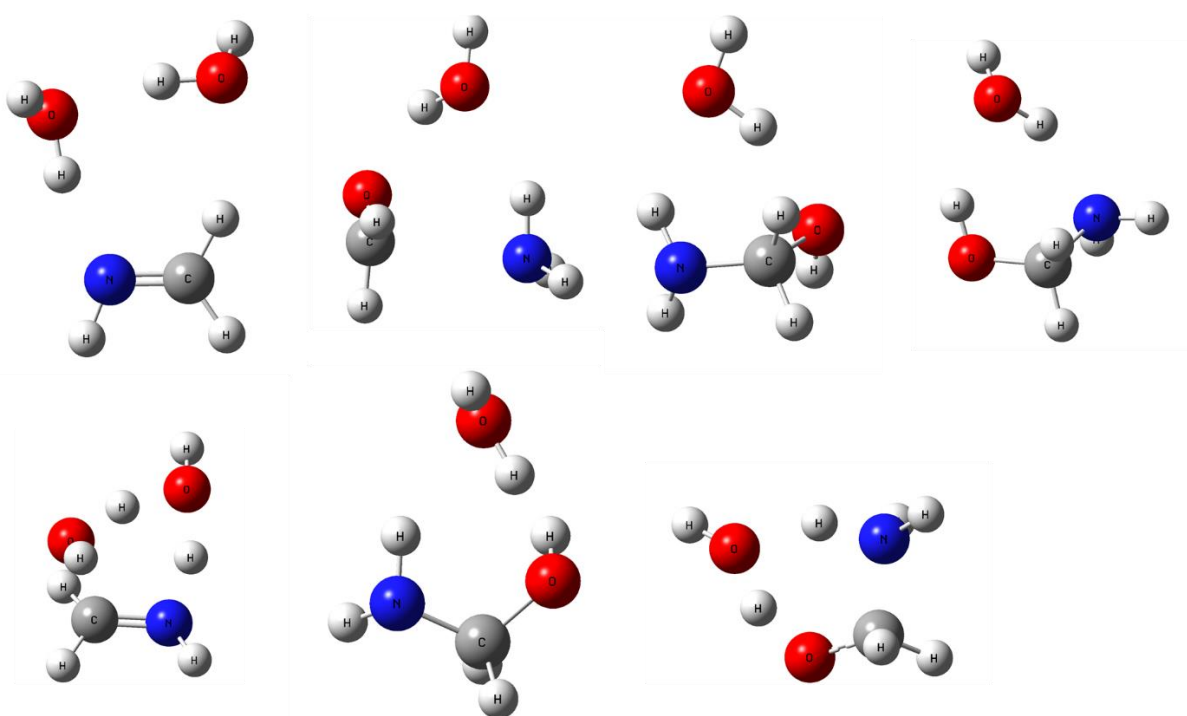


Figure S1: Optimized structure of reactants, complex, intermediates, products and transition states.

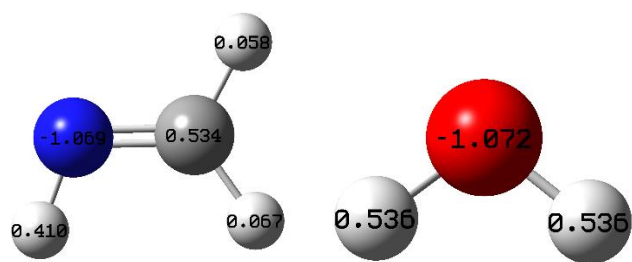


Figure S2: Calculated partial electric charge on each atom in CH_2NH and H_2O .

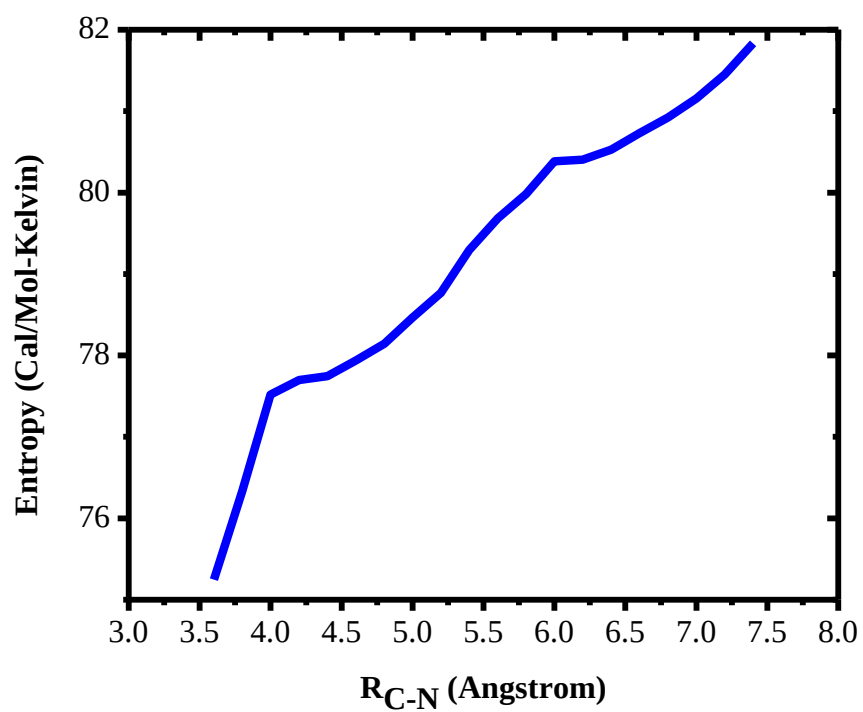


Figure S3: Entropy for dissociation of $\text{CH}_2\text{NH}\cdots\text{H}_2\text{O}$ as functions of N-O bond distance at 298K.

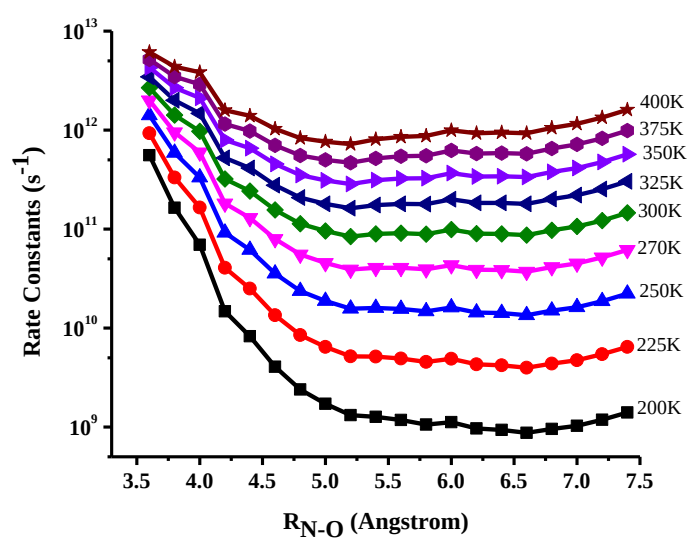


Figure S4: Trial rate constants for dissociation of $\text{CH}_2\text{NH}\cdots\text{H}_2\text{O}$ as functions of N-O bond distance from 200 to 400 K in the interval of 25K.

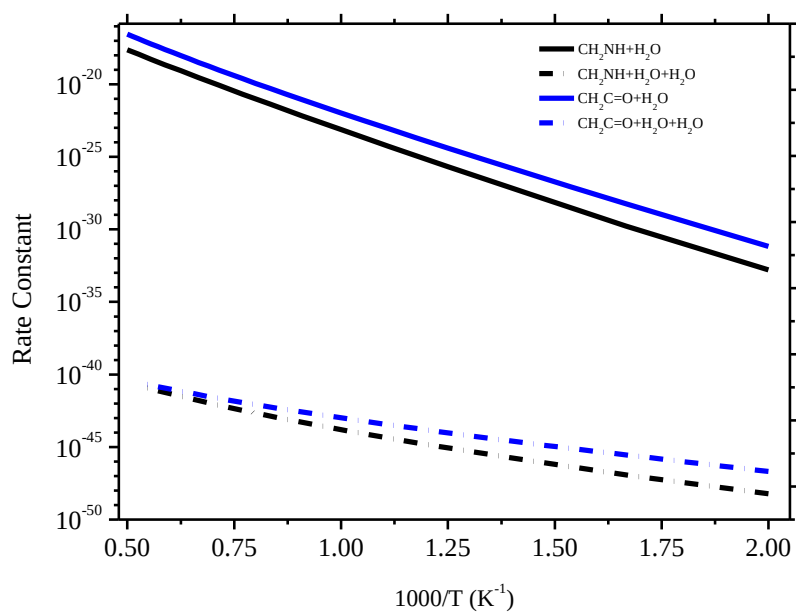


Figure S5: Comparison of rate coefficients (a) solid line_Bimolecular ($\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$): $\text{CH}_2\text{NH}+\text{H}_2\text{O}$ with $\text{CH}_2\text{C}=\text{O}+\text{H}_2\text{O}$ (b) Dash line_ Termolecular Reaction ($\text{cm}^6 \text{ molecule}^{-2} \text{ s}^{-1}$): $\text{CH}_2\text{NH}+\text{H}_2\text{O}+\text{H}_2\text{O}$ with $\text{CH}_2\text{C}=\text{O}+\text{H}_2\text{O}+\text{H}_2\text{O}$.

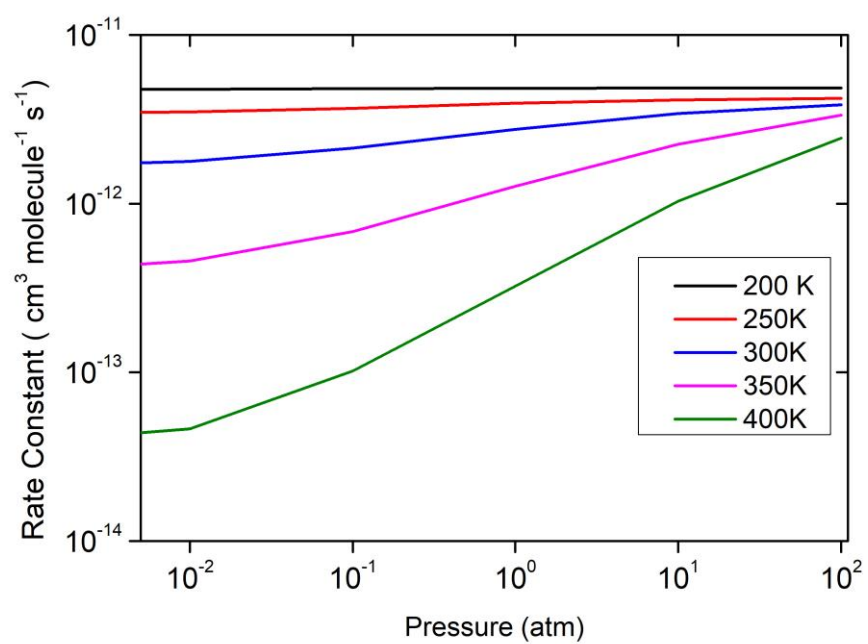


Figure.S6 Temperature-and pressure-dependent rate constants for $\text{CH}_2\text{NH} + \text{H}_2\text{O} \rightarrow \text{H}_2\text{O} \cdots \text{CH}_2\text{NH} \cdots \text{H}_2\text{O}$.

Details of Theoretical Calculations:

The rate constants were calculated using dual dynamic level with the interpolated single-point energies (ISPE).⁴⁰ The minimum energy pathway is obtained using direct dynamics approach for a small range of the reaction path with the mass scaled reaction coordinate from -1.5 to 1.5 Bohr by using the Page–McIver integrator with a step size of 0.05 Bohr. The bimolecular rate constants for one-water reaction were calculated using Eq.S1:

$$k(T)_\infty = k^{CVT}(T) \times K_{eq}(T) \quad (S1)$$

The equilibrium constants $K_{eq}(T)$ were calculated by EqS2:

$$K_{eq}(T) = \frac{Q_{CH_2NH \cdots H_2O}}{Q_{CH_2NH} \times Q_{H_2O}} \times \exp\left(-\frac{E_0^{CH_2NH \cdots H_2O} - E_0^{CH_2NH} - E_0^{H_2O}}{k_B T}\right). \quad (S2)$$

Where $Q_{CH_2NH \cdots H_2O}$, Q_{CH_2NH} and Q_{H_2O} are total partition functions for pre-reactive complex, methylenimine and water, and E_0 is the zero-point corrected energies of complex and reactants.

The bimolecular rate constants for two-water reaction were calculated using Eq.S3:

$$k(T, H_2O)_\infty = k_{H_2O}^{CVT}(T) \times K_{eq}(T, H_2O) \quad (3)$$

The equilibrium constants $K_{eq}(T, H_2O)$ for the formation of complexes were calculated using Eq.S4 and EqS5:

$$K_{eq}^1(T, H_2O) = \frac{Q_{PRC-H_2O}}{Q_{CH_2NH-H_2O} \times Q_{H_2O}} \times \exp\left(-\frac{\Delta E_0^{(PRC-H_2O)_1}}{k_B T}\right). \quad (S4)$$

$$K_{eq}^2(T, H_2O) = \frac{Q_{PRC-H_2O}}{Q_{H_2O-H_2O} \times Q_{CH_2NH}} \times \exp\left(-\frac{\Delta E_0^{(PRC-H_2O)_2}}{k_B T}\right). \quad (S5)$$

where Q_{PRC-H_2O} , $Q_{CH_2NH-H_2O}$, $Q_{H_2O-H_2O}$ are total partition functions for different complexes, , and ΔE_0 is the zero-point corrected energies difference between pre-reactive complex and reactants.

The pressure-dependent bimolecular rate constants for $CH_2NH + H_2O$ was calculated based on ref.

$$k_{tot}^{rec}(T, M) = K_{eq} k_\infty^{uni} f_{CH_2NH \cdots H_2O} \quad (S6)$$

where $k_\infty^{uni}(T)$ is the high-pressure limit rate constant for $CH_2NH \cdots H_2O \rightarrow CH_2NH + H_2O$,

K_{eq} is the equilibrium constant and $f_{CH_2NH \cdots H_2O}$ is the calculated fractional of

$CH_2NH \cdots H_2O$ during simulations consisted of 10^6 stochastic trials

As shown in PES (Fig. 3), a kinetic scheme can be summarized as $\text{CH}_2\text{NH} + \text{H}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons [\text{CH}_2\text{NH}\cdots\text{H}_2\text{O}\cdots\text{H}_2\text{O}] \Rightarrow \text{Products}$ And a rate constant (k_p) to form products can be computed as:

$k_p = K_{eq} \times k_f$ for tri-molecular reaction (See Figure S4)

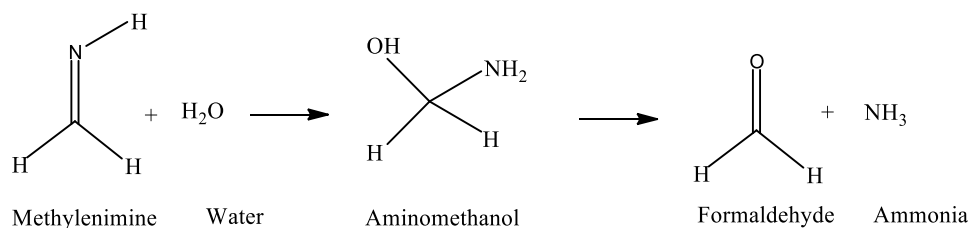
or $k_p = K_{eq} \times k_f \times [\text{H}_2\text{O}]$ for pseudo bi-molecular reaction

where K_{eq} is the thermal equilibrium constant for the $\text{CH}_2\text{NH} + \text{H}_2\text{O} + \text{H}_2\text{O} \rightleftharpoons [\text{CH}_2\text{NH}\cdots\text{H}_2\text{O}\cdots\text{H}_2\text{O}]$ that is calculated using THERMO code, and k_f is the rate constant for the $[\text{CH}_2\text{NH}\cdots\text{H}_2\text{O}\cdots\text{H}_2\text{O}] \Rightarrow \text{Products}$ (from CVT/SCT), which is extremely slow and the rate-determining step. The K_{eq} is completely independent on a number of pathways (e.g. Pathways A and B) and how it can be reached. The equilibrium (K_{eq}) depends only on the starting-point and the end-point according to thermodynamic regulations.

Therefore, we believe, the kinetic scheme of $\text{CH}_2\text{NH} + \text{H}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{NH}\cdots\text{H}_2\text{O}\cdots\text{H}_2\text{O} \rightarrow \text{Products}$. The rate-determining step is at the step 2, which is the same in both pathways. Therefore, the correct equation to calculate the rate constant from reactants to products is expressed by Eq. 10

$$k_p = k_A^{eff} = k_B^{eff}$$

Scheme1: Reaction mechanism of methylenimine with water molecule to produce formaldehyde and ammonia *via* aminomethanol.



THERMO Input File

(1) $\text{H}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{H}_2\text{O} \cdots \text{H}_2\text{O}$

KCAL MCC

25

200. 225. 250. 275. 300. 325. 350. 375. 400. 500. 600. 700. 800. 900. 1000. 1100. 1200. 1300.
1400. 1500. 1600. 1700. 1800. 1900. 2000.

3

reac water 0.0 ! ATcT v.1.118a

H2O

! NIST-JANAF 1998

!

! small revisions based Bernath & Colin, JMS 257, 20-23 (2009)

2 1 1

0.0 1

5 'HAR' 'GHZ'

1 vib 1619.24 0 1

2 vib 3889.6 0 1

3 vib 3990.6 0 1

4 qro 830.14341 1 1 ! mo6

5 qro 350.9942744 1 2 ! mo6

reac water 0.0 ! ATcT v.1.118a

H2O

! NIST-JANAF 1998

!

! small revisions based Bernath & Colin, JMS 257, 20-23 (2009)

2 1 1

0.0 1

5 'HAR' 'GHZ'

1 vib 1619.24 0 1

2 vib 3889.6 0 1

3 vib 3990.6 0 1

4 qro 830.14341 1 1 ! mo6

5 qro 350.9942744 1 2 ! mo6

prod H2O--H2O_____ -2.9

H2OH2O

a) NIST / JANAF 1998

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

| | | | | |
|----|-------|-------------|---|-----|
| 14 | 'HAR' | 'GHZ' | | |
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| 2 | vib | 172.65 | 0 | 1 |
| 3 | vib | 190.29 | 0 | 1 |
| 4 | vib | 207.41 | 0 | 1 |
| 5 | vib | 378.62 | 0 | 1 |
| 6 | vib | 635.48 | 0 | 1 |
| 7 | vib | 1622.83 | 0 | 1 |
| 8 | vib | 1638.71 | 0 | 1 |
| 9 | vib | 3799.08 | 0 | 1 |
| 10 | vib | 3885.07 | 0 | 1 |
| 11 | vib | 3965.18 | 0 | 1 |
| 12 | vib | 3982.55 | 0 | 1 |
| 13 | qro | 214.9604 | | 1 1 |
| 14 | qro | 6.506562637 | | 1 2 |

(2) CH₂NH + H₂O → CH₂NH---H₂O

KCAL MCC

25

200. 225. 250. 275. 300. 325. 350. 375. 400. 500. 600. 700. 800. 900. 1000. 1100. 1200. 1300.
1400. 1500. 1600. 1700. 1800. 1900. 2000.

3

reac ch2nh_____ 0.0

CH2NH

a) Frequencies from NIST Web-Book (11/2002)

b) Moments of Inertia from JANAF (1998)

c) delta-H(0 K) from JANAF (1998)

1 1 1

0.0 1

| | | | | |
|----|-------|------------|--|-----|
| 11 | 'HAR' | 'GHZ' | | |
| 1 | vib | 1076.990 | | 1 |
| 2 | vib | 1122.380 | | 1 |
| 3 | vib | 1169.680 | | 1 |
| 4 | vib | 1370.410 | | 1 |
| 5 | vib | 1501.130 | | 1 |
| 6 | vib | 1744.990 | | 1 |
| 7 | vib | 3068.770 | | 1 |
| 8 | vib | 3153.020 | | 1 |
| 9 | vib | 3471.720 | | 1 |
| 10 | qro | 198.84379 | | 1 1 |
| 11 | qro | 32.5443753 | | 1 2 |

reac water 0.0 ! ATcT v.1.118a

H2O

! NIST-JANAF 1998

!

! small revisions based Bernath & Colin, JMS 257, 20-23 (2009)

2 1 1
 0.0 1
 5 'HAR' 'GHZ'
 1 vib 1619.24 0 1
 2 vib 3889.6 0 1
 3 vib 3990.6 0 1
 4 qro 830.14341 1 1 ! mo6
 5 qro 350.9942744 1 2 ! mo6

prod CH₂NH--H₂O_____ -4.4
 CH₂NHH₂O

a) NIST / JANAF 1998

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

20 'HAR' 'GHZ'

| | | | | |
|----|-----|-------------|---|---|
| 1 | vib | 118.71 | 0 | 1 |
| 2 | vib | 165.88 | 0 | 1 |
| 3 | vib | 208.14 | 0 | 1 |
| 4 | vib | 209.94 | 0 | 1 |
| 5 | vib | 371.3 | 0 | 1 |
| 6 | vib | 618.42 | 0 | 1 |
| 7 | vib | 1081.490 | | 1 |
| 8 | vib | 1132.410 | | 1 |
| 9 | vib | 1158.3 | 0 | 1 |
| 10 | vib | 1378.330 | | 1 |
| 11 | vib | 1502.990 | | 1 |
| 12 | vib | 1642.320 | | 1 |
| 13 | vib | 1749.620 | | 1 |
| 14 | vib | 3076.340 | | 1 |
| 15 | vib | 3177.940 | | 1 |
| 16 | vib | 3496.440 | | 1 |
| 17 | vib | 3719.140 | | 1 |
| 18 | vib | 3971.110 | | 1 |
| 19 | qro | 34.27515 | 1 | 1 |
| 20 | qro | 4.696331507 | 1 | 2 |

(3) H₂O...H₂O + CH₂NH → PRC

KCAL MCC

25

200. 225. 250. 275. 300. 325. 350. 375. 400. 500. 600. 700. 800. 900. 1000. 1100.
 1200. 1300. 1400. 1500. 1600. 1700. 1800. 1900. 2000.

3

reac H₂O--H₂O_____ 0.0

H₂OH₂O

a) NIST / JANAF 1998

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

14 'HAR' 'GHZ'

| | | | | |
|----|-----|-------------|---|---|
| 1 | vib | 159.5 0 | 1 | |
| 2 | vib | 172.65 0 | 1 | |
| 3 | vib | 190.29 0 | 1 | |
| 4 | vib | 207.41 0 | 1 | |
| 5 | vib | 378.62 0 | 1 | |
| 6 | vib | 635.48 0 | 1 | |
| 7 | vib | 1622.83 | 0 | 1 |
| 8 | vib | 1638.71 | 0 | 1 |
| 9 | vib | 3799.08 | 0 | 1 |
| 10 | vib | 3885.07 | 0 | 1 |
| 11 | vib | 3965.18 | 0 | 1 |
| 12 | vib | 3982.55 | 0 | 1 |
| 13 | qro | 214.9604 | 1 | 1 |
| 14 | qro | 6.506562637 | 1 | 2 |

reac ch2nh_____ 0.0

CH2NH

a) Frequencies from NIST Web-Book (11/2002)

b) Moments of Inertia from JANAF (1998)

c) delta-H(0 K) from JANAF (1998)

1 1 1

0.0 1

11 'HAR' 'GHZ'

| | | | | |
|----|-----|------------|---|---|
| 1 | vib | 1076.99 0 | 1 | |
| 2 | vib | 1122.38 0 | 1 | |
| 3 | vib | 1169.68 0 | 1 | |
| 4 | vib | 1370.41 0 | 1 | |
| 5 | vib | 1501.13 0 | 1 | |
| 6 | vib | 1744.99 0 | 1 | |
| 7 | vib | 3068.77 0 | 1 | |
| 8 | vib | 3153.02 0 | 1 | |
| 9 | vib | 3471.72 0 | 1 | |
| 10 | qro | 198.84379 | 1 | 1 |
| 11 | qro | 32.5443753 | 1 | 2 |

prod CH2NH--2H2O_____ -8.60

CH2NHH4O2

a) NIST / JANAF 1998

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

29 'HAR' 'GHZ'

| | | | | |
|----|-----|-------------|---|---|
| 1 | vib | 64.38 | 0 | 1 |
| 2 | vib | 78.61 | 0 | 1 |
| 3 | vib | 164.37 | 0 | 1 |
| 4 | vib | 193.83 | 0 | 1 |
| 5 | vib | 211.46 | 0 | 1 |
| 6 | vib | 219.67 | 0 | 1 |
| 7 | vib | 233.19 | 0 | 1 |
| 8 | vib | 240.45 | 0 | 1 |
| 9 | vib | 412.69 | 0 | 1 |
| 10 | vib | 441.28 | 0 | 1 |
| 11 | vib | 704.08 | 0 | 1 |
| 12 | vib | 843.17 | 0 | 1 |
| 13 | vib | 1105.89 | 0 | 1 |
| 14 | vib | 1146.34 | 0 | 1 |
| 15 | vib | 1171.71 | 0 | 1 |
| 16 | vib | 1394.03 | 0 | 1 |
| 17 | vib | 1514.09 | 0 | 1 |
| 18 | vib | 1647.34 | 0 | 1 |
| 19 | vib | 1660.69 | 0 | 1 |
| 20 | vib | 1743.46 | 0 | 1 |
| 21 | vib | 3070.29 | 0 | 1 |
| 22 | vib | 3185.22 | 0 | 1 |
| 23 | vib | 3485.08 | 0 | 1 |
| 24 | vib | 3500.81 | 0 | 1 |
| 25 | vib | 3647.65 | 0 | 1 |
| 26 | vib | 3952.34 | 0 | 1 |
| 27 | vib | 3970.63 | 0 | 1 |
| 28 | qro | 6.13309 | 1 | 1 |
| 29 | qro | 2.697160156 | 1 | 2 |

(4) $\text{CH}_2\text{NH}\cdots\text{H}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{PRC}$

KCAL MCC

25

200. 225. 250. 275. 300. 325. 350. 375. 400. 500. 600. 700. 800. 900. 1000. 1100.
1200. 1300. 1400. 1500. 1600. 1700. 1800. 1900. 2000.

3

```

reac  CH2NH--H2O_____ 0.0
CH2NHH2O
a) NIST / JANAF 1998
b) (blank comment line)
c) (blank comment line)
1  1  1
0.0 1
20 'HAR' 'GHZ'
1  vib  118.71 0  1
2  vib  165.88 0  1
3  vib  208.14 0  1
4  vib  209.94 0  1
5  vib  371.3  0  1
6  vib  618.42 0  1
7  vib  1081.49 0  1
8  vib  1132.41 0  1
9  vib  1158.3  0  1
10 vib  1378.33 0  1
11 vib  1502.99 0  1
12 vib  1642.32 0  1
13 vib  1749.62 0  1
14 vib  3076.34 0  1
15 vib  3177.94 0  1
16 vib  3496.44 0  1
17 vib  3719.14 0  1
18 vib  3971.11 0  1
19 qro  34.27515  1  1
20 qro  4.696331507  1  2

```

```

reac water 0.0 ! ATcT v.1.118a
H2O
! NIST-JANAF 1998
!
! small revisions based Bernath & Colin, JMS 257, 20-23 (2009)
2 1 1
0.0 1
5 'HAR' 'GHZ'
1 vib  1619.24  0  1
2 vib  3889.6   0  1
3 vib  3990.6   0  1
4 qro  830.14341  1  1 ! mo6
5 qro  350.9942744  1  2 ! mo6

```

prod CH2NH--2H2O_____ -7.2
CH2NHH4O2

a) NIST / JANAF 1998

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

29 'HAR' 'GHZ'

| | | | | |
|----|-----|-------------|---|---|
| 1 | vib | 64.38 | 0 | 1 |
| 2 | vib | 78.61 | 0 | 1 |
| 3 | vib | 164.37 | 0 | 1 |
| 4 | vib | 193.83 | 0 | 1 |
| 5 | vib | 211.46 | 0 | 1 |
| 6 | vib | 219.67 | 0 | 1 |
| 7 | vib | 233.19 | 0 | 1 |
| 8 | vib | 240.45 | 0 | 1 |
| 9 | vib | 412.69 | 0 | 1 |
| 10 | vib | 441.28 | 0 | 1 |
| 11 | vib | 704.08 | 0 | 1 |
| 12 | vib | 843.17 | 0 | 1 |
| 13 | vib | 1105.89 | 0 | 1 |
| 14 | vib | 1146.34 | 0 | 1 |
| 15 | vib | 1171.71 | 0 | 1 |
| 16 | vib | 1394.03 | 0 | 1 |
| 17 | vib | 1514.09 | 0 | 1 |
| 18 | vib | 1647.34 | 0 | 1 |
| 19 | vib | 1660.69 | 0 | 1 |
| 20 | vib | 1743.46 | 0 | 1 |
| 21 | vib | 3070.29 | 0 | 1 |
| 22 | vib | 3185.22 | 0 | 1 |
| 23 | vib | 3485.08 | 0 | 1 |
| 24 | vib | 3500.81 | 0 | 1 |
| 25 | vib | 3647.65 | 0 | 1 |
| 26 | vib | 3952.34 | 0 | 1 |
| 27 | vib | 3970.63 | 0 | 1 |
| 28 | qro | 6.13309 | 1 | 1 |
| 29 | qro | 2.697160156 | 1 | 2 |

KTOOLS Input File

CH2NH+H2O -> CH2O+NH3 ro-harmonic vib model M062X/pople enthalpies

KCAL MCC ! Ekey, Sskey
 both ! Whatdo
 85000 10. ! Emax Egrain
 500 1 ! Jmax Jgrain
 500 5000 85000. ! 1D double array parameters: Imax1, Isize, Emax2
 9 ! Nt number of temperatures
 200. 225. 250. 275. 300. 325. 350. 375. 400.
 1 20 2 ! no. of reactants, no. of of trial transition states, no. of products

reac CH2NH-H2O 0.0 0.0

CH2NHH2O

a) NIST / JANAF 2003, Dorofeeva et al. J. Phys. Chem. Ref. Data 32 (2003) 879

b) (blank comment line)

c) (blank comment line)

1 1 1

0.0 1

20 'HAR' 'GHZ'

| | | | | |
|----|-----|-------------|---|---|
| 1 | vib | 118.71 | 0 | 1 |
| 2 | vib | 165.88 | 0 | 1 |
| 3 | vib | 208.14 | 0 | 1 |
| 4 | vib | 209.94 | 0 | 1 |
| 5 | vib | 371.3 | 0 | 1 |
| 6 | vib | 618.42 | 0 | 1 |
| 7 | vib | 1081.49 | 0 | 1 |
| 8 | vib | 1132.41 | 0 | 1 |
| 9 | vib | 1158.3 | 0 | 1 |
| 10 | vib | 1378.33 | 0 | 1 |
| 11 | vib | 1502.99 | 0 | 1 |
| 12 | vib | 1642.32 | 0 | 1 |
| 13 | vib | 1749.62 | 0 | 1 |
| 14 | vib | 3076.34 | 0 | 1 |
| 15 | vib | 3177.94 | 0 | 1 |
| 16 | vib | 3496.44 | 0 | 1 |
| 17 | vib | 3719.14 | 0 | 1 |
| 18 | vib | 3971.11 | 0 | 1 |
| 19 | kro | 34.27515 | 1 | 1 |
| 20 | jro | 4.696331507 | 1 | 2 |

ctst CH2NH-H2O-36 1.33 3.6 0

CH2NHH2O

a) Akbar

b) m062x

| c) | constrained | optimization | (OH-NO2 | distance | constrained) |
|----|-------------|--------------|---------|----------|--------------|
| 1 | 1 | 1 | | | |
| 0 | 1 | | | | |
| 19 | 'HAR' | 'GHZ' | | | |
| 1 | vib | 59.79 0 | 1 | | |
| 2 | vib | 79.21 0 | 1 | | |
| 3 | vib | 188.08 0 | 1 | | |
| 4 | vib | 243.44 0 | 1 | | |
| 5 | vib | 467.44 0 | 1 | | |
| 6 | vib | 1087.09 | 0 | 1 | |
| 7 | vib | 1121.84 | 0 | 1 | |
| 8 | vib | 1166.9 0 | 1 | | |
| 9 | vib | 1310.87 | 0 | 1 | |
| 10 | vib | 1495.28 | 0 | 1 | |
| 11 | vib | 1630.43 | 0 | 1 | |
| 12 | vib | 1727.08 | 0 | 1 | |
| 13 | vib | 3062.84 | 0 | 1 | |
| 14 | vib | 3158.59 | 0 | 1 | |
| 15 | vib | 3469.53 | 0 | 1 | |
| 16 | vib | 3794.74 | 0 | 1 | |
| 17 | vib | 3957.12 | 0 | 1 | |
| 18 | kro | 42.50561 | 1 | 1 | ! GHz |
| 19 | jro | 2.757894573 | 1 | 2 | ! GHz |

ctst CH2NH-H2O-38 1.95 3.8 0
CH2NHH2O

| a) | Akbar | | | | |
|----|-------------|--------------|---------|----------|--------------|
| b) | m062x | | | | |
| c) | constrained | optimization | (OH-NO2 | distance | constrained) |
| 1 | 1 | 1 | | | |
| 0 | 1 | | | | |
| 19 | 'HAR' | 'GHZ' | | | |
| 1 | vib | 60.22 0 | 1 | | |
| 2 | vib | 67.41 0 | 1 | | |
| 3 | vib | 159.54 0 | 1 | | |
| 4 | vib | 223.58 0 | 1 | | |
| 5 | vib | 421.67 0 | 1 | | |
| 6 | vib | 1085.57 | 0 | 1 | |
| 7 | vib | 1120.81 | 0 | 1 | |
| 8 | vib | 1167.79 | 0 | 1 | |
| 9 | vib | 1308.42 | 0 | 1 | |
| 10 | vib | 1495.01 | 0 | 1 | |
| 11 | vib | 1627.43 | 0 | 1 | |
| 12 | vib | 1725.21 | 0 | 1 | |
| 13 | vib | 3062.45 | 0 | 1 | |

| | | | | | | |
|----|-----|-------------|---|---|---|-----|
| 14 | vib | 3157.85 | 0 | 1 | | |
| 15 | vib | 3467.98 | 0 | 1 | | |
| 16 | vib | 3818.43 | 0 | 1 | | |
| 17 | vib | 3954.26 | 0 | 1 | | |
| 18 | kro | 43.1247 | 1 | 1 | ! | GHz |
| 19 | jro | 2.499388937 | 1 | 2 | ! | GHz |

ctst CH2NH-H2O-40 2.44 4 0

CH2NHH2O

a) Akbar

b) m062x

c) constrained optimization (ch2nh-h2o distance constrained)

1 1 1

0 1

19 'HAR' 'GHZ'

1 vib 57.28 0 1

2 vib 63.01 0 1

3 vib 126.47 0 1

4 vib 203.09 0 1

5 vib 377.94 0 1

6 vib 1084.44 0 1

7 vib 1120.54 0 1

8 vib 1168.69 0 1

9 vib 1306.69 0 1

10 vib 1494.86 0 1

11 vib 1625.39 0 1

12 vib 1723.79 0 1

13 vib 3063.15 0 1

14 vib 3158.01 0 1

15 vib 3464.99 0 1

16 vib 3835.43 0 1

17 vib 3952.58 0 1

18 kro 43.28132 1 1 ! GHz

19 jro 2.282428917 1 2 ! GHz

ctst CH2NH-H2O-42 2.99 4.2 0

CH2NHH2O

a) Nguyen, Barker,et al. (2011+)

b) UB97-1/m062x

c) constrained optimization (OH-NO2 distance constrained)

1 1 1

0 1

19 'HAR' 'GHZ'

1 vib 62.54 0 1

| | | | | | | |
|----|-----|-------------|---|---|---|-----|
| 2 | vib | 77.09 0 | 1 | | | |
| 3 | vib | 133.83 0 | 1 | | | |
| 4 | vib | 208.64 0 | 1 | | | |
| 5 | vib | 346.51 0 | 1 | | | |
| 6 | vib | 1083.94 | 0 | 1 | | |
| 7 | vib | 1123.32 | 0 | 1 | | |
| 8 | vib | 1170.16 | 0 | 1 | | |
| 9 | vib | 1306.35 | 0 | 1 | | |
| 10 | vib | 1495.57 | 0 | 1 | | |
| 11 | vib | 1619.11 | 0 | 1 | | |
| 12 | vib | 1722.62 | 0 | 1 | | |
| 13 | vib | 3067.4 0 | 1 | | | |
| 14 | vib | 3164.6 0 | 1 | | | |
| 15 | vib | 3454.93 | 0 | 1 | | |
| 16 | vib | 3847.95 | 0 | 1 | | |
| 17 | vib | 3961.94 | 0 | 1 | | |
| 18 | kro | 43.29473 | 1 | 1 | ! | GHz |
| 19 | jro | 2.093259836 | 1 | 2 | ! | GHz |

| | | | | | | |
|----------|--------------|--------------|---------|----------|--------------|-----|
| ctst | CH2NH-H2O-44 | 3.3 | 4.4 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Nguyen, | Barker,et | al. | (2011+) | | |
| b) | UB97-1/m062x | | | | | |
| c) | constrained | optimization | (OH-NO2 | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 60.07 0 | 1 | | | |
| 2 | vib | 73.52 0 | 1 | | | |
| 3 | vib | 142.06 0 | 1 | | | |
| 4 | vib | 187.35 0 | 1 | | | |
| 5 | vib | 302.04 0 | 1 | | | |
| 6 | vib | 1083.37 | 0 | 1 | | |
| 7 | vib | 1122.64 | 0 | 1 | | |
| 8 | vib | 1170.57 | 0 | 1 | | |
| 9 | vib | 1305.7 0 | 1 | | | |
| 10 | vib | 1495.47 | 0 | 1 | | |
| 11 | vib | 1618.4 0 | 1 | | | |
| 12 | vib | 1722.53 | 0 | 1 | | |
| 13 | vib | 3066.82 | 0 | 1 | | |
| 14 | vib | 3163.48 | 0 | 1 | | |
| 15 | vib | 3453.84 | 0 | 1 | | |
| 16 | vib | 3848.22 | 0 | 1 | | |
| 17 | vib | 3965.5 0 | 1 | | | |
| 18 | kro | 42.9707 | 1 | 1 | ! | GHz |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|-----|
| 19 | jro | 1.931292032 | 1 | 2 | ! | GHz |
| | | | | | | |
| ctst | CH2NH-H2O-46 | 3.61 | 4.6 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Nguyen, | Barker,et | al. | (2011+) | | |
| b) | UB97-1/m062x | | | | | |
| c) | constrained | optimization | (OH-NO2 | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 58.7 | 0 | 1 | | |
| 2 | vib | 74.1 | 0 | 1 | | |
| 3 | vib | 159.4 | 0 | 1 | | |
| 4 | vib | 183.08 | 0 | 1 | | |
| 5 | vib | 272.72 | 0 | 1 | | |
| 6 | vib | 1083.16 | 0 | 1 | | |
| 7 | vib | 1122.37 | 0 | 1 | | |
| 8 | vib | 1170.92 | 0 | 1 | | |
| 9 | vib | 1305.64 | 0 | 1 | | |
| 10 | vib | 1495.84 | 0 | 1 | | |
| 11 | vib | 1618.55 | 0 | 1 | | |
| 12 | vib | 1722.84 | 0 | 1 | | |
| 13 | vib | 3066.65 | 0 | 1 | | |
| 14 | vib | 3163.37 | 0 | 1 | | |
| 15 | vib | 3452.03 | 0 | 1 | | |
| 16 | vib | 3851.51 | 0 | 1 | | |
| 17 | vib | 3967.58 | 0 | 1 | | |
| 18 | kro | 42.64997 | 1 | 1 | ! | GHz |
| 19 | jro | 1.786914092 | 1 | 2 | ! | GHz |
| | | | | | | |
| ctst | CH2NH-H2O-48 | 3.85 | 4.8 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 57.39 | 0 | 1 | | |
| 2 | vib | 79.4 | 0 | 1 | | |
| 3 | vib | 157.75 | 0 | 1 | | |
| 4 | vib | 182.17 | 0 | 1 | | |
| 5 | vib | 255.61 | 0 | 1 | | |
| 6 | vib | 1083.05 | 0 | 1 | | |
| 7 | vib | 1122.61 | 0 | 1 | | |
| 8 | vib | 1171.39 | 0 | 1 | | |

| | | | | | | |
|----|-----|-------------|---|---|---|-----|
| 9 | vib | 1305.32 | 0 | 1 | | |
| 10 | vib | 1496.3 0 | 1 | | | |
| 11 | vib | 1617.06 | 0 | 1 | | |
| 12 | vib | 1722.88 | 0 | 1 | | |
| 13 | vib | 3066.83 | 0 | 1 | | |
| 14 | vib | 3164.5 0 | 1 | | | |
| 15 | vib | 3449.35 | 0 | 1 | | |
| 16 | vib | 3855.05 | 0 | 1 | | |
| 17 | vib | 3969.31 | 0 | 1 | | |
| 18 | kro | 42.40552 | 1 | 1 | ! | GHz |
| 19 | jro | 1.657221088 | 1 | 2 | ! | GHz |

ctst CH2NH-H2O-50 4.03 5 0

CH2NHH2O

a) Akbar

b) m062x

c) constrained optimization ch2nh-h2o distance constrained)

1 1 1

0 1

19 'HAR' 'GHZ'

1 vib 55.82 0 1

2 vib 81.76 0 1

3 vib 151.7 0 1

4 vib 181.85 0 1

5 vib 241.48 0 1

6 vib 1082.56 0 1

7 vib 1122.04 0 1

8 vib 1171.72 0 1

9 vib 1303.65 0 1

10 vib 1496.42 0 1

11 vib 1616.33 0 1

12 vib 1722.18 0 1

13 vib 3066.46 0 1

14 vib 3165.14 0 1

15 vib 3447.01 0 1

16 vib 3858.11 0 1

17 vib 3969.64 0 1

18 kro 42.16518 1 1 ! GHz

19 jro 1.541015543 1 2 ! GHz

ctst CH2NH-H2O-52 4.18 5.2 0

CH2NHH2O

a) Akbar

b) m062x

c) constrained optimization ch2nh-h2o distance constrained)

1 1 1

| | | | | | | |
|----|-------|-------------|---|---|---|-----|
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 54.71 0 | 1 | | | |
| 2 | vib | 83.13 0 | 1 | | | |
| 3 | vib | 146.45 0 | 1 | | | |
| 4 | vib | 180.68 0 | 1 | | | |
| 5 | vib | 232.06 0 | 1 | | | |
| 6 | vib | 1082.77 | 0 | 1 | | |
| 7 | vib | 1121.34 | 0 | 1 | | |
| 8 | vib | 1171.93 | 0 | 1 | | |
| 9 | vib | 1306.28 | 0 | 1 | | |
| 10 | vib | 1497.18 | 0 | 1 | | |
| 11 | vib | 1614.5 0 | 1 | | | |
| 12 | vib | 1723.33 | 0 | 1 | | |
| 13 | vib | 3065.76 | 0 | 1 | | |
| 14 | vib | 3165.7 0 | 1 | | | |
| 15 | vib | 3444.38 | 0 | 1 | | |
| 16 | vib | 3861.36 | 0 | 1 | | |
| 17 | vib | 3970.55 | 0 | 1 | | |
| 18 | kro | 41.9142 | 1 | 1 | ! | GHz |
| 19 | jro | 1.436604909 | 1 | 2 | ! | GHz |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|--|
| ctst | CH2NH-H2O-54 | 4.27 | 5.4 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 52.98 0 | 1 | | | |
| 2 | vib | 83.8 0 | 1 | | | |
| 3 | vib | 128.45 0 | 1 | | | |
| 4 | vib | 177.82 0 | 1 | | | |
| 5 | vib | 225.4 0 | 1 | | | |
| 6 | vib | 1082.65 | 0 | 1 | | |
| 7 | vib | 1119.43 | 0 | 1 | | |
| 8 | vib | 1172.1 0 | 1 | | | |
| 9 | vib | 1306.77 | 0 | 1 | | |
| 10 | vib | 1497.8 0 | 1 | | | |
| 11 | vib | 1611.87 | 0 | 1 | | |
| 12 | vib | 1723.67 | 0 | 1 | | |
| 13 | vib | 3064.72 | 0 | 1 | | |
| 14 | vib | 3166.23 | 0 | 1 | | |
| 15 | vib | 3441.03 | 0 | 1 | | |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|-----|
| 16 | vib | 3863.01 | 0 | 1 | | |
| 17 | vib | 3971.99 | 0 | 1 | | |
| 18 | kro | 41.64897 | 1 | 1 | ! | GHz |
| 19 | jro | 1.342497332 | 1 | 2 | ! | GHz |
| | | | | | | |
| ctst | CH2NH-H2O-56 | | 4.36 | 5.6 | 0 | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 51.41 0 | 1 | | | |
| 2 | vib | 84.18 0 | 1 | | | |
| 3 | vib | 119.54 0 | 1 | | | |
| 4 | vib | 175.65 0 | 1 | | | |
| 5 | vib | 220.25 0 | 1 | | | |
| 6 | vib | 1082.52 | 0 | 1 | | |
| 7 | vib | 1116.92 | 0 | 1 | | |
| 8 | vib | 1172.21 | 0 | 1 | | |
| 9 | vib | 1306.17 | 0 | 1 | | |
| 10 | vib | 1498.23 | 0 | 1 | | |
| 11 | vib | 1610.52 | 0 | 1 | | |
| 12 | vib | 1723.78 | 0 | 1 | | |
| 13 | vib | 3063.86 | 0 | 1 | | |
| 14 | vib | 3166.72 | 0 | 1 | | |
| 15 | vib | 3438.93 | 0 | 1 | | |
| 16 | vib | 3864.5 0 | 1 | | | |
| 17 | vib | 3973.06 | 0 | 1 | | |
| 18 | kro | 41.48547 | 1 | 1 | ! | GHz |
| 19 | jro | 1.256634808 | 1 | 2 | ! | GHz |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|--|
| ctst | CH2NH-H2O-58 | | 4.45 | 5.8 | 0 | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 50.01 0 | 1 | | | |
| 2 | vib | 84.24 0 | 1 | | | |
| 3 | vib | 115.84 0 | 1 | | | |
| 4 | vib | 174.45 0 | 1 | | | |

| | | | | | | |
|----|-----|-------------|---|---|---|-----|
| 5 | vib | 215.24 0 | 1 | | | |
| 6 | vib | 1082.41 | 0 | 1 | | |
| 7 | vib | 1111.9 0 | 1 | | | |
| 8 | vib | 1172.58 | 0 | 1 | | |
| 9 | vib | 1305.08 | 0 | 1 | | |
| 10 | vib | 1499.55 | 0 | 1 | | |
| 11 | vib | 1608.34 | 0 | 1 | | |
| 12 | vib | 1724.09 | 0 | 1 | | |
| 13 | vib | 3062.81 | 0 | 1 | | |
| 14 | vib | 3167.16 | 0 | 1 | | |
| 15 | vib | 3437.09 | 0 | 1 | | |
| 16 | vib | 3874.79 | 0 | 1 | | |
| 17 | vib | 3979.19 | 0 | 1 | | |
| 18 | kro | 41.33573 | 1 | 1 | ! | GHz |
| 19 | jro | 1.178617706 | 1 | 2 | ! | GHz |

ctst CH2NH-H2O-60 4.49 6 0

CH2NHH2O

a) Akbar

b) m062x

c) constrained optimization ch2nh-h2o distance constrained)

1 1 1

0 1

19 'HAR' 'GHZ'

1 vib 48.54 0 1

2 vib 82.61 0 1

3 vib 108.42 0 1

4 vib 173.36 0 1

5 vib 211.07 0 1

6 vib 1082.51 0 1

7 vib 1107.68 0 1

8 vib 1172.08 0 1

9 vib 1308.38 0 1

10 vib 1499.46 0 1

11 vib 1608.09 0 1

12 vib 1724.73 0 1

13 vib 3061.76 0 1

14 vib 3167.31 0 1

15 vib 3434.41 0 1

16 vib 3868.81 0 1

17 vib 3976.9 0 1

18 kro 41.1221 1 1 ! GHz

19 jro 1.107966872 1 2 ! GHz

ctst CH2NH-H2O-62 4.56 6.2 0

CH2NHH2O

| | | | | | | |
|----|-------------|--------------|-----------|----------|--------------|-----|
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 47.73 0 | 1 | | | |
| 2 | vib | 81.65 0 | 1 | | | |
| 3 | vib | 118.91 0 | 1 | | | |
| 4 | vib | 174.43 0 | 1 | | | |
| 5 | vib | 207.74 0 | 1 | | | |
| 6 | vib | 1081.8 0 | 1 | | | |
| 7 | vib | 1101.98 | 0 | 1 | | |
| 8 | vib | 1171.97 | 0 | 1 | | |
| 9 | vib | 1307.59 | 0 | 1 | | |
| 10 | vib | 1499.93 | 0 | 1 | | |
| 11 | vib | 1607.57 | 0 | 1 | | |
| 12 | vib | 1725.07 | 0 | 1 | | |
| 13 | vib | 3060.92 | 0 | 1 | | |
| 14 | vib | 3167.43 | 0 | 1 | | |
| 15 | vib | 3433.68 | 0 | 1 | | |
| 16 | vib | 3870.15 | 0 | 1 | | |
| 17 | vib | 3979.67 | 0 | 1 | | |
| 18 | kro | 40.99231 | 1 | 1 | ! | GHz |
| 19 | jro | 1.043033628 | 1 | 2 | ! | GHz |

ctst CH2NH-H2O-64 4.6 6.4 0
CH2NHH2O

| | | | | | | |
|----|-------------|--------------|-----------|----------|--------------|--|
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 46.4 0 | 1 | | | |
| 2 | vib | 79.96 0 | 1 | | | |
| 3 | vib | 126.48 0 | 1 | | | |
| 4 | vib | 175.1 0 | 1 | | | |
| 5 | vib | 204.15 0 | 1 | | | |
| 6 | vib | 1082.21 | 0 | 1 | | |
| 7 | vib | 1092.96 | 0 | 1 | | |
| 8 | vib | 1171.47 | 0 | 1 | | |
| 9 | vib | 1311.38 | 0 | 1 | | |
| 10 | vib | 1500.75 | 0 | 1 | | |
| 11 | vib | 1607.14 | 0 | 1 | | |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|-----|
| 12 | vib | 1726.11 | 0 | 1 | | |
| 13 | vib | 3060.08 | 0 | 1 | | |
| 14 | vib | 3167.55 | 0 | 1 | | |
| 15 | vib | 3430.83 | 0 | 1 | | |
| 16 | vib | 3868.25 | 0 | 1 | | |
| 17 | vib | 3978.98 | 0 | 1 | | |
| 18 | kro | 40.82333 | 1 | 1 | ! | GHz |
| 19 | jro | 0.983798054 | 1 | 2 | ! | GHz |
| | | | | | | |
| ctst | CH2NH-H2O-66 | 4.64 | 6.6 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 45.12 0 | 1 | | | |
| 2 | vib | 78.33 0 | 1 | | | |
| 3 | vib | 130.32 0 | 1 | | | |
| 4 | vib | 174.49 0 | 1 | | | |
| 5 | vib | 200.24 0 | 1 | | | |
| 6 | vib | 1078.68 | 0 | 1 | | |
| 7 | vib | 1085.16 | 0 | 1 | | |
| 8 | vib | 1171.16 | 0 | 1 | | |
| 9 | vib | 1313.08 | 0 | 1 | | |
| 10 | vib | 1501.25 | 0 | 1 | | |
| 11 | vib | 1607.17 | 0 | 1 | | |
| 12 | vib | 1727.19 | 0 | 1 | | |
| 13 | vib | 3059.41 | 0 | 1 | | |
| 14 | vib | 3167.66 | 0 | 1 | | |
| 15 | vib | 3428.7 0 | 1 | | | |
| 16 | vib | 3870.92 | 0 | 1 | | |
| 17 | vib | 3981.93 | 0 | 1 | | |
| 18 | kro | 40.68325 | 1 | 1 | ! | GHz |
| 19 | jro | 0.983798054 | 1 | 2 | ! | GHz |
| | | | | | | |
| ctst | CH2NH-H2O-68 | 4.66 | 6.8 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 43.92 0 | 1 | | | |

| | | | | | | |
|----|-----|-------------|---|---|---|-----|
| 2 | vib | 76.95 | 0 | 1 | | |
| 3 | vib | 134.99 | 0 | 1 | | |
| 4 | vib | 172.67 | 0 | 1 | | |
| 5 | vib | 196.56 | 0 | 1 | | |
| 6 | vib | 1068.62 | 0 | 1 | | |
| 7 | vib | 1083.44 | 0 | 1 | | |
| 8 | vib | 1170.77 | 0 | 1 | | |
| 9 | vib | 1315.35 | 0 | 1 | | |
| 10 | vib | 1501.93 | 0 | 1 | | |
| 11 | vib | 1607.65 | 0 | 1 | | |
| 12 | vib | 1727.85 | 0 | 1 | | |
| 13 | vib | 3058.65 | 0 | 1 | | |
| 14 | vib | 3167.78 | 0 | 1 | | |
| 15 | vib | 3426.61 | 0 | 1 | | |
| 16 | vib | 3872.31 | 0 | 1 | | |
| 17 | vib | 3982.94 | 0 | 1 | | |
| 18 | kro | 40.56842 | 1 | 1 | ! | GHz |
| 19 | jro | 0.879162502 | 1 | 2 | ! | GHz |

ctst CH2NH-H2O-70 4.67 7.0 0

CH2NHH2O

a) Akbar

b) m062x

c) constrained optimization ch2nh-h2o distance constrained)

1 1 1

0 1

19 'HAR' 'GHZ'

1 vib 42.81 0 1

2 vib 75.29 0 1

3 vib 138.55 0 1

4 vib 168.53 0 1

5 vib 193.14 0 1

6 vib 1048.16 0 1

7 vib 1084.63 0 1

8 vib 1170.22 0 1

9 vib 1307.45 0 1

10 vib 1501.05 0 1

11 vib 1622.35 0 1

12 vib 1723.03 0 1

13 vib 3059 0 1

14 vib 3167.87 0 1

15 vib 3412.87 0 1

16 vib 3832.3 0 1

17 vib 3953.4 0 1

18 kro 40.45894 1 1 ! GHz

19 jro 0.832907923 1 2 ! GHz

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|-----|
| ctst | CH2NH-H2O-72 | 4.66 | 7.2 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 41.95 0 | 1 | | | |
| 2 | vib | 73.7 0 | 1 | | | |
| 3 | vib | 140.72 0 | 1 | | | |
| 4 | vib | 159.33 0 | 1 | | | |
| 5 | vib | 190.78 0 | 1 | | | |
| 6 | vib | 1037.3 0 | 1 | | | |
| 7 | vib | 1083.15 | 0 | 1 | | |
| 8 | vib | 1169.97 | 0 | 1 | | |
| 9 | vib | 1319.52 | 0 | 1 | | |
| 10 | vib | 1502.87 | 0 | 1 | | |
| 11 | vib | 1608.28 | 0 | 1 | | |
| 12 | vib | 1730.16 | 0 | 1 | | |
| 13 | vib | 3057.87 | 0 | 1 | | |
| 14 | vib | 3167.92 | 0 | 1 | | |
| 15 | vib | 3422.29 | 0 | 1 | | |
| 16 | vib | 3871.59 | 0 | 1 | | |
| 17 | vib | 3984.62 | 0 | 1 | | |
| 18 | kro | 40.35742 | 1 | 1 | ! | GHz |
| 19 | jro | 0.790167478 | 1 | 2 | ! | GHz |

| | | | | | | |
|----------|--------------|--------------|-----------|----------|--------------|--|
| ctst | CH2NH-H2O-74 | 4.65 | 7.4 | 0 | | |
| CH2NHH2O | | | | | | |
| a) | Akbar | | | | | |
| b) | m062x | | | | | |
| c) | constrained | optimization | ch2nh-h2o | distance | constrained) | |
| 1 | 1 | 1 | | | | |
| 0 | 1 | | | | | |
| 19 | 'HAR' | 'GHZ' | | | | |
| 1 | vib | 41.04 0 | 1 | | | |
| 2 | vib | 72.01 0 | 1 | | | |
| 3 | vib | 141.91 0 | 1 | | | |
| 4 | vib | 144.47 0 | 1 | | | |
| 5 | vib | 189.28 0 | 1 | | | |
| 6 | vib | 1021.6 0 | 1 | | | |
| 7 | vib | 1083.1 0 | 1 | | | |
| 8 | vib | 1169.61 | 0 | 1 | | |
| 9 | vib | 1323.25 | 0 | 1 | | |

| | | | | | | |
|----|-----|-----------|---|---|---|-----|
| 10 | vib | 1503.41 | 0 | 1 | | |
| 11 | vib | 1607.91 | 0 | 1 | | |
| 12 | vib | 1731.66 | 0 | 1 | | |
| 13 | vib | 3057.32 | 0 | 1 | | |
| 14 | vib | 3167.95 | 0 | 1 | | |
| 15 | vib | 3420.75 | 0 | 1 | | |
| 16 | vib | 3870.61 | 0 | 1 | | |
| 17 | vib | 3984.71 | 0 | 1 | | |
| 18 | kro | 40.2442 | 1 | 1 | ! | GHz |
| 19 | jro | 0.7506513 | 1 | 2 | ! | GHz |

prod ch2nh 4.4 0.0

CH2NH

a) Akbar

b) Calculated

c) from Akbar

1 1 1

0.0 1

11 'HAR' 'GHZ'

1 vib 1076.99 0 1

2 vib 1122.38 0 1

3 vib 1169.68 0 1

4 vib 1370.41 0 1

5 vib 1501.13 0 1

6 vib 1744.99 0 1

7 vib 3068.77 0 1

8 vib 3153.02 0 1

9 vib 3471.72 0 1

10 kro 198.84379 1 1

11 jro 32.5443753 1 2

prod water 0.0 0.0

H2O

a) Akbar

b) Calculated

c) from Akbar

2 1 1

0.0 1

5 'HAR' 'GHZ'

1 vib 1619.24 0 1

2 vib 3889.6 0 1

3 vib 3990.6 0 1

4 kro 830.14341 1 1 ! mo6

5 jro 350.9942744 1 2 ! mo6